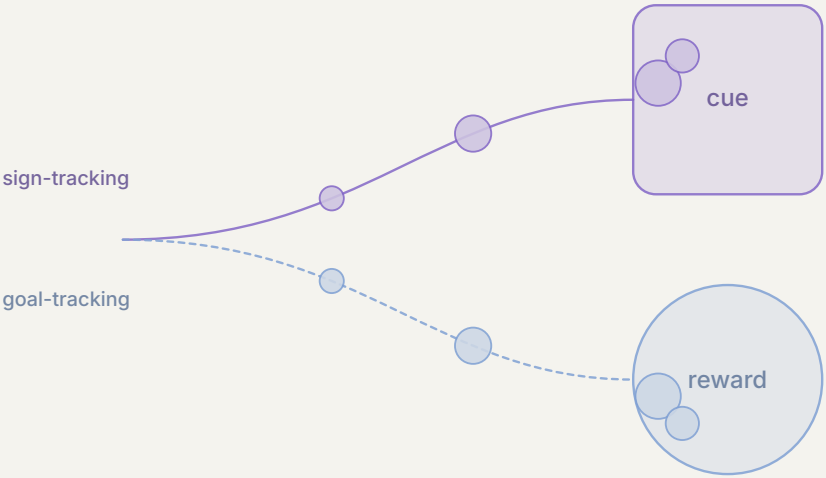
WORKSHOP • PROGRAMME & ABSTRACTS

Sign- and Goal-Tracking

Human data, computational models, and rodent foundations

DATE
20–22 November 2026

VENUE
Potsdam, Germany



FUNDED BY

SFB 1294 Data Assimilation

Universität Potsdam

HMU Health and Medical University

WELCOME

Dear colleagues and friends

Research on individual differences in Pavlovian conditioned approach has developed rapidly: from animal work on the mechanisms behind sign- and goal-tracking, to computational accounts, to human studies using eye-tracking, approach tasks and neural measures.

Although these approaches increasingly inform one another, researchers working with the different paradigms and levels of analysis rarely meet in the same room. This workshop brings them together to discuss how sign- and goal-tracking are currently conceptualised, measured, and modelled.

Lever autoshaping, gaze-based measures, neural measures, physical approach tasks and formal computational models each capture different aspects of the phenotype. By discussing them side by side, we hope to see where the lines of work converge, where they differ, and which open questions matter most for future research. The sessions emphasise discussion and exchange, with time to explain not only findings but also the practical details of how the work is done.

Daniel J. Schad
Organiser • Health and Medical University Potsdam

Ralf Engbert
Organiser • University of Potsdam

SPEAKERS

Our speakers

Patrick Anselme
Ruhr Universität Bochum, Bochum, Germany

Romana Caposova
HMU Health and Medical University, Potsdam, Germany

François Cinotti
Sorbonne Université, Paris, France

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Sara Garofalo
University of Bologna, Bologna, Italy

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Reinout Wiers
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Laura Wirth
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OVERVIEW

Contents

Welcome 2

Speakers 3

Programme 5

Day 1 • Friday, 20th November 5

Day 2 • Saturday, 21st November 6

Day 3 • Sunday, 22nd November 7

Practical information 8

Talks 10

Posters 23

Poster abstracts 25

Funding 38

HOW TO READ THIS BOOKLET

T1 Talk P1 Poster

Presenting authors are set in **bold**. Page numbers beside each talk point to its abstract. Talks marked (online) are given remotely.

PROGRAMME • ANIMAL FOUNDATIONS

Day 1 • Friday, 20th November

14:00–15:00	Registration
15:00–15:30	Welcome & intro Daniel J. Schad
15:30–16:30 p. 10	**Leveraging Individual Differences in Cue-Motivated Behavior to Identify Neural Mechanisms of Inhibitory Control** Shelly Flagel (online), University of Michigan
16:30–17:00	*Coffee break*
17:00–18:00 p. 11	**Sign-Tracking $\neq$ Sign-Trackers** Terry Robinson (online), University of Michigan
18:00–18:30 p. 11	**Individual differences and neural circuit mechanisms of cue-guided learning** Benjamin Saunders (online), University of Minnesota
18:30–19:00 p. 12	**Does reward uncertainty induce aversive sign-tracking?** Patrick Anselme, Ruhr Universität Bochum
19:30	Workshop dinner at Zeppelin Restaurant, Kongresshotel Potsdam

PROGRAMME • COMPUTATIONAL MODELLING AND HUMAN DATA

Day 2 • Saturday, 21st November

09:00–09:45 p. 12
Sign-tracking and goal-tracking behaviours as an interplay between model-based and model-free learning processes
François Cinotti, Sorbonne Université

09:45–10:45 p. 13
HeiDI: From general processes to individual differences
Robert Honey, Cardiff University

10:45–11:15 *Coffee break*

11:15–12:15 p. 13
Tracking the trackers: evidence, measurement, and open challenges from a decade of human sign-tracking and goal-tracking research
Sara Garofalo, University of Bologna

12:15–13:15 *Lunch break*

13:15–14:15 p. 14
Dissociating neural learning signals in human sign- and goal-trackers
Daniel J. Schad, HMU Health and Medical University

14:15–15:15 p. 14
When the Signal Becomes the Target: Sign-Tracking and the Brain's Response to Reward
Martin Paulus, Laureate Institute for Brain Research

15:15–15:45 *Coffee break*

15:45–16:45 p. 15
Human sign- and goal-tracking: From Basic Mechanisms to Translational Challenges
Michelle Heck, University of Liège

16:45–17:15 p. 16
Substance-related Cognitive Biases in Humans, Sign-Tracking?
Reinout Wiers (online), University of Amsterdam

17:15–18:45 POSTERS
Poster session (P1–P13)

PROGRAMME • MORE DATA, INCLUDING RODENTS AND HUMANS

Day 3 • Sunday, 22nd November

09:00–09:30 p. 17
Beyond Incentive Salience: Sign-Tracking, Goal-Tracking, and Contextual Control
Jonathan Morrow, University of Michigan

09:30–10:00 p. 17
Sign-Tracking as a Transdiagnostic Trait: Thalamic Circuits Linking Reward and Fear Processing
Cristina Orsini, Sapienza University of Rome

10:00–10:30 p. 18
Reproducible human reward imaging phenotypes exhibit differential sensitivity to dopamine D2 receptor antagonism
Nicola Sambuco, University of Bari Aldo Moro

10:30–11:00 *Coffee break*

11:00–12:00 p. 19
From chocolate to cigarettes: Sign-tracking-like brain responses to food cues predict cue-induced eating and smoking lapse
Francesco Versace, The University of Texas

12:00–12:30 p. 20
Do human sign- and goal-trackers differ in model-free and model-based behavior, and how can we identify them at scale?
Laura Wirth, HMU Health and Medical University

12:30–13:30 *Lunch break*

13:30–13:45 p. 21
Enhanced Pavlovian-to-instrumental transfer in human sign-trackers: bias or slower updating?
Romana Caposova, HMU Health and Medical University

13:45–14:15 p. 21
Sign- and goal-tracking in human alcohol dependence
Nassim Sadedin, HMU Health and Medical University

14:15–14:30 Concluding remarks
Daniel J. Schad

PRACTICAL INFORMATION

Venue & logistics

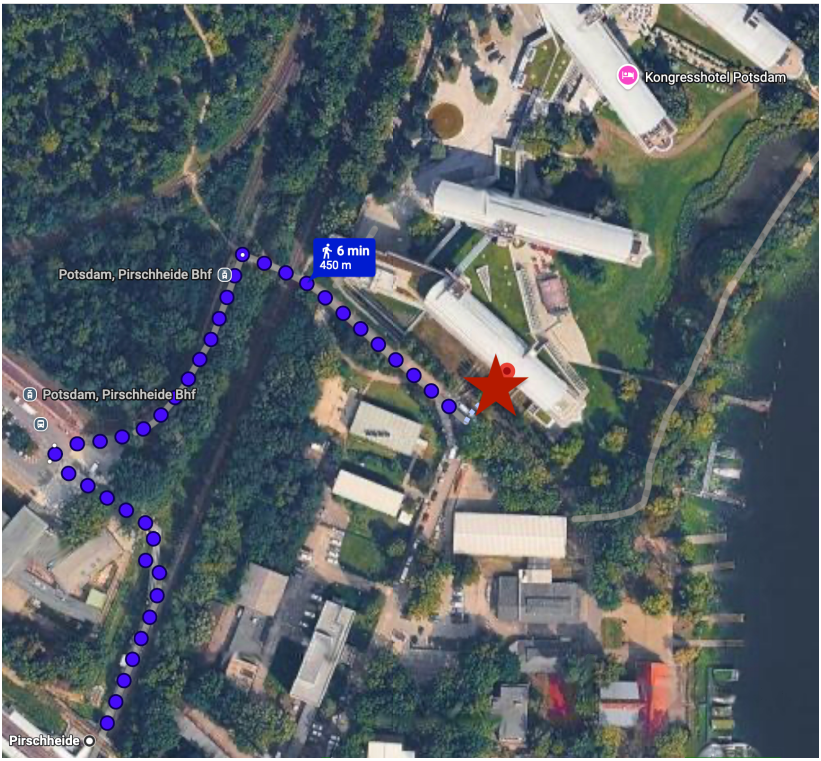
VENUE	Lecture hall (Hörsaal) A2.06, close to the Kongresshotel Potsdam, Am Luftschiffhafen, Potsdam. The red star on the map marks the entrance of the workshop building.
GETTING THERE	Use the train station Potsdam Pirschheide to get there. From the airport (BER), there are direct trains to Potsdam Pirschheide (RB22, running hourly). Please check maps.google.com or int.bahn.de for connections. See the map for details.
REGISTRATION	20th November, 14:00–15:00, at the venue.
POSTERS	Poster size: up to A0 (portrait or landscape). Mounting material is provided.
CONTACT	stgtworkshop.potsdam@gmail.com

Online participation

The Zoom link and the link for questions during the talks are sent to registered participants by email. To join online, please write to stgtworkshop.potsdam@gmail.com.

VENUE

Map



The map shows the way from the train station Pirschheide (lower left corner) to the venue (red star); 6 min walking distance. The hotel (Kongresshotel Potsdam) is visible in the top right.

Entrance at 52°22′27.1″N 13°00′50.5″E

Map data © Google. Red star: workshop building entrance.

ABSTRACTS

Talks

T1 Leveraging Individual Differences in Cue-Motivated Behavior to Identify Neural Mechanisms of Inhibitory Control

Shelly Flagel

University of Michigan, Ann Arbor, USA

Individuals vary in how they respond to cues in their surrounding environment. This variability reflects differences in learned associations and the value attributed to environmental cues. Studies in rodents reveal inherent individual differences in how animals learn about and respond to reward-predictive cues. When rats are exposed to a Pavlovian conditioned approach procedure, in which a lever cue is repeatedly paired with food reward, some rats—known as sign-trackers—attribute incentive motivational value (i.e., incentive salience) to the cue and, in turn, approach and interact with it upon its presentation. In contrast, goal-trackers approach the location of impending reward delivery upon cue presentation. Both sign- and goal-trackers learn the cue–reward relationship, but the motivational value they attribute to the cue differs fundamentally. Relative to goal-trackers, sign-trackers exhibit greater impulsivity and compulsivity, deficits in attentional control, and greater cue-induced reinstatement of drug-seeking behavior (i.e., relapse). These behavioral phenotypes also diverge neurobiologically: goal-trackers rely on cortical control systems, whereas sign-trackers show greater reliance on subcortical emotional and motivational circuits. Thus, individual variation in Pavlovian conditioned approach behavior provides an animal model of neurobehavioral endophenotypes relevant to psychiatric disorders. The identification of analogous individual differences across species provides a translational framework for investigating shared behavioral and neural mechanisms as well as the factors that influence them. In this talk, I will present published and unpublished cross-species data that build on the animal foundations of the sign-tracker/goal-tracker model and illustrate how individual differences in cue-motivated behavior can be leveraged to uncover neural mechanisms underlying deficits in inhibitory control characteristic of substance use disorders and other disorders of impulse control.

T2 Sign-Tracking ≠ Sign-Trackers

Terry Robinson

University of Michigan, Ann Arbor, USA

Rats described as sign-trackers (STs) vs. goal-trackers (GTs) are usually identified by their propensity to develop a sign-tracking or goal-tracking conditioned response (ST CR or GT CR) when a lever-CS is paired with a food US. However, acquisition of a ST or GT CR is only a single feature of what are very complex neurobehavioral phenotypes. For example, a cue associated with food or drug rewards is also a more effective conditioned reinforcer and more effective in evoking a conditioned motivational state in STs than GTs, which is one reason we have suggested conditioned emotional/motivational states are important contributors to these neurobehavioral phenotypes. In this talk I will focus on several less well known, less appreciated and less well understood features of STs and GTs that need to be considered in developing computational models of these neurobehavioral phenotypes or designing parallel studies in humans. For example, there is considerable evidence that dopamine is necessary for the performance of a ST CR but not a GT CR, when a lever is used as the CS, but the form of the CS can very much influence both the nature of the CR and its neural underpinnings. In addition, some of the largest differences between STs and GTs are in attentional/executive control over behavior and associated differences in prefrontal cortical acetylcholine neurotransmission. STs are also more impulsive and show more dysfunctional checking behavior than GTs. The implications of these and other features of STs and GTs for the development of computational models will be discussed, including the notion that dopamine is in fact not required for learning a CS (lever)-US association, even in STs.

T3 Individual differences and neural circuit mechanisms of cue-guided learning

Benjamin Saunders

University of Minnesota, Minneapolis, USA

Environmental cues can promote orientation and approach behavior, directing seeking behaviors into close proximity with reward. Critically, there are substantial individual differences in the extent that reward cues promote approach and attentional bias, pointing to different paths of vulnerability to motivational disorders. In this talk, I will discuss past studies using a rodent model of individual differences in Pavlovian conditioned approach, highlighting distinct behavioral strategies biased toward cues

(sign tracking) versus goals (goal tracking), and the role of striatal dopamine. Then, I will talk about newer unpublished work investigating the parafascicular thalamus, a region that projects strongly to the striatum, as a brain locus of cue-related behavioral control. Our studies suggest that this thalamic region is important for steering attention to salient environmental cues, to promote exploration and learning, and may therefore be a promising target to understand individual differences in approach behavior.

T4 Does reward uncertainty induce aversive sign-tracking?

Patrick Anselme

Ruhr Universität Bochum, Bochum, Germany

Most research on sign-tracking has focused on appetitive approach behavior, in which the unconditioned stimulus is a reward. A few autoshaping studies have nevertheless provided evidence of aversive sign-tracking in rats, although these effects appear less pronounced than those observed in appetitive contexts. Given the tendency of animals and humans to avoid uncertainty in the absence of potential gains, such uncertainty may be experienced as predominantly aversive. Yet, in autoshaping, reward uncertainty has been shown to increase the speed of approach to the conditioned stimulus, as well as to induce more vigorous, persistent, and/or variable responding in rats, pigeons, and quails. I suggest that these responses reflect a form of aversive sign-tracking: an incentive process allowing organisms to counter the difficulty that uncertainty represents, rather than one driven primarily by appetitive attraction to the uncertain reward. Numerous examples of behaviors that may illustrate this process can be found in natural settings.

T5 Sign-tracking and goal-tracking behaviours as an interplay between model-based and model-free learning processes

François Cinotti

Sorbonne Université, Paris, France

Confronted with a common Pavlovian conditioning experience in which a lever predicts delivery of a reward, individual animals adopt strikingly distinct behaviours, known as sign-tracking, if they engage the predictive lever, and goal-tracking, if they prefer attending to the site of future reward delivery. In this talk, I will present a unifying computational framework which combines model-based decision-making, that relies on an internal representation of the task structure, with attribution of incentive salience to environmental stimuli. After describing the mechanisms of this model, I shall overview

its success in fitting and explaining behavioural, physiological, and pharmacological data, before discussing predictions made by the model, in particular how to bias a group of animals towards goal-tracking, as well as implications for behaviour beyond Pavlovian conditioning.

T6 HeiDI: From general processes to individual differences

Robert Honey

Cardiff University, Cardiff, UK

I describe the development and application of HeiDI, a computational model of Pavlovian conditioning originally inspired by findings from autoshaping procedures in rats (e.g., Flagel, Akil & Robinson, 2009). Building on the formal model of Rescorla and Wagner (1972) and the ideas of Asratian (1965), HeiDI provides a unified account of a broad range of phenomena in Pavlovian conditioning (Honey, Dwyer & Iliescu, 2020; Navarro, Dwyer & Honey, 2023). Importantly, HeiDI bridges the gap between general associative processes and individual differences expressed as sign-tracking and goal-tracking behaviours.

T7 Tracking the trackers: evidence, measurement, and open challenges from a decade of human sign-tracking and goal-tracking research

Sara Garofalo

University of Bologna, Bologna, Italy

The distinction between sign-trackers (ST) and goal-trackers (GT) captures individual differences in how reward-predictive cues are processed and used to guide behaviour. This talk will retrace our group's work on bringing this distinction from animal to human research, from our first attempt to identify human STs and GTs to our most recent psychometric developments. It will begin by introducing our eye-tracking paradigm, followed by behavioural, computational, and electrophysiological evidence that sign-trackers are more susceptible to maladaptive Pavlovian biases in decision-making. Computational modelling traces this bias to slower updating of cue values in sign-trackers, which are also characterized by a neural marker of weaker cortical control over motivational processes. The talk will then take a critical look at the measure used to distinguish STs from GTs, presenting work that tests the reliability and validity of current tools and introduces a new version of the task. Taken together, these findings

establish ST/GT as a promising framework for human research, and they set out the methodological steps needed to turn it into a reliable tool for the clinical and non-clinical investigation of interindividual differences sensitivity to reward cues.

T8 Dissociating neural learning signals in human sign- and goal-trackers

Daniel J. Schad

HMU Health and Medical University, Potsdam, Germany

Individuals differ in how they learn from experience. In Pavlovian conditioning paradigms, where cues predict reinforcer delivery at a different goal location, some animals—so-called sign-trackers—come to approach the cue, whereas others, called goal-trackers, approach the goal. In sign-trackers, model-free phasic dopaminergic reward prediction errors underlie learning, which renders stimuli 'wanted'. Goal-trackers do not rely on dopamine for learning and are thought to use model-based learning. We demonstrate this double dissociation in 128 male humans using eye-tracking, pupillometry and fMRI informed by computational models of sign- and goal-tracking. We show that sign-trackers exhibit a neural reward prediction error signal that is not detectable in goal-trackers. Model-free value only guides gaze and pupil dilation in sign-trackers. Goal-trackers instead exhibit a stronger model-based neural state prediction error signal. This model-based construct determines gaze and pupil dilation more in goal-trackers.

T9 When the Signal Becomes the Target: Sign-Tracking and the Brain's Response to Reward

Martin P. Paulus

Laureate Institute for Brain Research, Tulsa, USA

Reward-predicting cues can direct behavior toward an anticipated outcome or become targets of approach themselves. The distinction between goal-tracking and sign-tracking provides a translational framework for investigating how predictive information acquires motivational significance. In youth, this framework offers an experimental approach to understanding individual differences in incentive processing and behavioral regulation.

This presentation integrates the LEGOS experiments with subsequent behavioral analyses, emphasizing the connection between cue-directed behavior and neuroimaging.

Forty children aged 9–12 years completed a response-independent conditioning task, and 29 underwent fMRI during a separate monetary incentive delay task. Greater sign-tracking was associated with weaker left parietal modulation during gain and loss anticipation and stronger right amygdala modulation during gain anticipation. Continuous brain–behavior correlations followed this regional pattern: $r = -0.46$ and -0.45 for parietal gain and loss contrasts, respectively, and $r = +0.48$ for amygdala gain. Subsequent behavioral analyses demonstrated strong consistency of cue-approach ranks across adjacent late-session blocks ($\rho = 0.82$). No distinct goal-tracking group emerged, highlighting the importance of examining cue approach dimensionally.

These findings motivate a central conceptual question: does attraction to reward cues reflect differences in attention, expectancy, action preparation, motivational value, or their interaction? The presentation considers how behavioral measurement and neuroimaging can distinguish these explanations while separating within-session consistency, neural association, and clinical prediction. The small imaging sample and selection of regions through related same-sample effects require independent replication. The broader objective is to develop cue approach into a precise experimental phenotype that connects observable behavior with neural incentive processing and supports targeted tests of mechanism.

T10 Human sign- and goal-tracking: From Basic Mechanisms to Translational Challenges

Michelle Heck

University of Liège, Liège, Belgium

Sign- and goal-tracking are reward cue-driven behaviors that have been extensively studied in animal models for decades. In contrast, the attempt to translate these two behavioral profiles to humans is relatively recent. From a translational perspective, however, it is crucial to determine whether findings from animal research generalize to humans and, ultimately, whether they provide meaningful insights into human behavior and psychopathology. A recent scoping review conducted by our team revealed that the emerging human sign-tracking literature is marked by substantial methodological heterogeneity and inconsistent findings. Consequently, several key questions remain unresolved. To what extent does the human analog capture the same underlying processes as those observed in animal models? How should sign-tracking be optimally implemented and measured in humans to fit animal models? And, importantly, how are the processes at stake relevant for the understanding of addiction vulnerability and

related outcomes in humans?

To address these questions, we conducted a series of studies using a computerized Pavlovian Conditioned Approach task designed to assess sign- and goal-tracking tendencies in humans. While our findings provide encouraging evidence for the feasibility and relevance of this framework, they also highlight a number of inconsistencies and unresolved issues. Taken together, our results suggest that important methodological and validity-related challenges have to be overcome. Addressing these challenges will be essential to determine the true translational value of the sign-tracker/goal-tracker framework and for realizing its potential contribution to human addiction research.

T11 Substance-related Cognitive Biases in Humans, Sign-Tracking?

Reinout Wiers

University of Amsterdam, Amsterdam, The Netherlands

In human research, two different classes of measurement paradigms can be distinguished, designed to assess cognitive biases related to substance use disorders (SUDs). A first class uses stimuli referring to the relevant class of substances, such as pictures of alcohol, with measures of attentional bias and approach bias being prime examples. We present some examples of paradigms (for which considerable improvement has been achieved in the past years regarding reliability) and findings with these paradigms.

The second class refers to paradigms in which previously neutral stimuli become associated with a reward (monetary or in some cases substance-related), such as varieties of the value-modulated attentional capture (VMAC) tasks. These tasks increase experimental control and share certain similarities with the original paradigms of sign-tracking, developed in animal research, as they assess reactions to signals of rewards (low vs. high reward) which achieved meaning during the task. Some meaningful correlations with substance use have been reported, although findings have not always been consistent. Important differences also remain, such as the fact that the meaning can be learned within the task or simply be told to the participant and that VMAC typically measures attentional capture rather than overt approach toward reward-predictive cues. We will show some relevant data for this class of paradigms.

Finally, there are recent tasks that combine features of both approaches, where an arbitrary signal predicts the subsequent occurrence of a SUD-relevant cue. We will also show some relevant data for this type of paradigm.

T12 Beyond Incentive Salience: Sign-Tracking, Goal-Tracking, and Contextual Control

Jonathan Morrow

University of Michigan, Ann Arbor, USA

Individuals differ markedly in how reward-predictive cues acquire control over attention, emotion, and motivation. Pavlovian conditioned approach distinguishes sign-trackers, which approach the predictive cue itself, from goal-trackers, which approach the location of impending reward. This talk will examine whether these phenotypes reflect broader differences in the information and learning processes that govern behavior. Our studies show that sign-tracking is more sensitive to outcome devaluation but less susceptible to blocking than goal-tracking, suggesting distinct contributions of incentive and predictive learning. These differences extend across valence: sign-trackers exhibit stronger fear responses to discrete aversive cues, whereas goal-trackers show greater contextual fear and context-controlled drug seeking. Human experiments reveal analogous variation in cue- versus outcome-directed gaze, with modest associations with impulsivity. Finally, chemogenetic studies implicate ventral hippocampus–nucleus accumbens circuitry in regulating conditioned approach and cued fear, with effects detected in males but not females, while ketamine reduces sign-tracking. Together, these findings support a framework in which PavCA reveals individual differences not only in incentive-salience attribution, but also in the relative control exerted by discrete cues and broader contextual information.

T13 Sign-Tracking as a Transdiagnostic Trait: Thalamic Circuits Linking Reward and Fear Processing

Cristina Orsini

Sapienza University of Rome, Rome, Italy

Sign-tracking (ST) is a behavioral phenotype in which reward-predictive cues acquire excessive motivational salience. Although not pathological per se, ST is associated with impulsivity and increased vulnerability to substance use disorders. Preclinical evidence further indicates that ST-prone individuals display a progressive, cue-triggered amplification of fear responses (fear incubation), modeling key features of anxiety-related conditions, including post-traumatic stress disorder. These observations support ST as a potential transdiagnostic trait linking maladaptive reward and fear processing. The paraventricular nucleus of the thalamus (PVT) is emerging as a critical hub co-

ordinating motivational and emotional states and may act as a functional "switch" between reward- and fear-related circuits. Using C57BL/6J (C57) and DBA/2J (DBA) male inbred mice, we previously showed that similar ST behavior may rely on distinct neural substrates, and that distinct PVT subregions contribute to ST development (posterior PVT in C57; anterior PVT in DBA). We now demonstrate that dopamine signaling within the nucleus accumbens differentially regulates ST expression across strains. Importantly, although both strains develop fear incubation, opposite PVT subregions contribute to development and expression of fear incubation (anterior PVT in C57; posterior PVT in DBA). Thus, similar behavioral phenotypes across genotypes can arise from distinct neural substrates, and the PVT circuits supporting ST are dissociable from those underlying fear incubation. These findings identify genotype-dependent circuit organization as a potential source of individual variability in vulnerability to addictive and anxiety-related disorders and provide a framework for investigating circuit-informed therapeutic strategies.

T14 Reproducible human reward imaging phenotypes exhibit differential sensitivity to dopamine D2 receptor antagonism

Nicola Sambuco

University of Bari Aldo Moro, Bari, Italy

Human reward processing varies along cue-centric and outcome-centric axes, but the mechanistic basis of individual variation in incentive salience attribution remains incompletely characterized. We show, using fMRI across five cohorts (N-total = 1251; N1 = 890; N2 = 245; N3 = 34; N4 = 48; N5 = 34), two reproducible imaging phenotypes mirroring sign- and goal-tracking (ST-like, GT-like). ST-like individuals showed dominant ventral striatal responses to reward-anticipation cues and sustained incentive salience attribution; GT-like individuals showed heightened responses to reward outcomes. This distinction was replicable across sites and independent samples. Single-dose and repeated-dose D2/D3 antagonism (risperidone, haloperidol, amisulpride) selectively reduced anticipatory ventral striatal activity in ST, with single-dose antagonism additionally producing a parallel drop in self-reported energy. Instead, D2/D3 partial agonism (aripiprazole) increased anticipatory and reduced outcome-phase responses in GT. In a psychosis cohort, antipsychotic D2 affinity was associated with blunted anticipatory signals and higher negative symptom burden, offering a neuroimaging-driven basis for stratifying patients and predicting response to dopaminergic agents.

T15 From chocolate to cigarettes: Sign-tracking-like brain responses to food cues predict cue-induced eating and smoking lapse

Francesco Versace

The University of Texas, Houston, USA

Drug-associated cues can trigger relapse in individuals with substance use disorders, yet not everyone exposed to them relapses. Individual differences in the tendency to attribute incentive salience to cues predicting rewards may partially explain this variability.

In animal models, this tendency is inferred from sign-tracking behavior. During Pavlovian conditioning, all animals learn that a cue predicts food delivery, but sign-trackers approach the cue itself, whereas goal-trackers approach the location where food is delivered. Sign-trackers are also more vulnerable to addiction-like behaviors. Compared with goal-trackers, they are more motivated to obtain drugs and show greater cue-induced reinstatement of drug seeking after extinction. Physiological states, such as deprivation, can exacerbate this vulnerability in sign-trackers.

We hypothesized that, in humans too, the tendency to attribute incentive salience to reward-predictive cues is a latent trait that influences behaviors across reward domains. We predicted that smokers attributing greater incentive salience to food-predictive cues would show greater cue-induced eating and earlier smoking in a laboratory lapse model, especially after nicotine deprivation.

We enrolled 103 adults who smoked $\geq$5 cigarettes/day. They completed two visits, one after smoking as usual and one after 24 hours of abstinence. At each visit, we recorded event-related potentials during the cued food availability task, followed by a validated laboratory model of smoking lapse. During the task, participants viewed pleasant, unpleasant, neutral, and food-related images. Food-related images signaled the availability of a chocolate candy that participants could eat or discard. We estimated the incentive salience of food-predictive cues relative to emotional images using the late positive potential (LPP), a robust index of motivational relevance. K-means clustering on Visit 1 LPPs identified two profiles: sign-tracking-like (STL; LPPs to food-predictive cues as large as those to emotional images) and goal-tracking-like (GTL; LPPs to food-predictive cues similar to those to neutral images). STL participants ate twice as many candies as GTL participants at both visits ($p < .001$). In the lapse model, the profiles did not differ after smoking as usual, but STL participants smoked sooner after abstinence ($p < .05$).

Neuroaffective responses to food-predictive cues uncover individual differences in incentive salience attribution that predict reward seeking across domains. Nicotine deprivation increases the behavioral expression of these differences, which may explain why drug cues trigger relapse only in some individuals. These profiles could help identify individuals most vulnerable to cue-induced relapse and guide personalized treatments.

T16 Do human sign- and goal-trackers differ in model-free and model-based behavior, and how can we identify them at scale?

Laura Wirth

HMU Health and Medical University, Potsdam, Germany

Individuals differ in how reward-predictive cues guide their behavior. Sign-trackers approach the cue itself, whereas goal-trackers approach the location of reward delivery. Neural evidence links sign-tracking to model-free and goal-tracking to model-based learning. We tested whether this dissociation is also evident in human behavior, and how sign- and goal-tracking can be measured at scale. In 200 participants, we combined gaze-based classification of sign- and goal-trackers with a two-stage Pavlovian-instrumental transfer (PIT) paradigm that separates model-free from model-based PIT. Overall, we found evidence for model-based PIT, presumably favored by detailed instructions and a rewarded query task, whereas evidence for model-free PIT was inconclusive. In individuals where query performance was low, sign-trackers exhibited more model-free and goal-trackers more model-based PIT, both statistically and in computational model parameters. Phenotype differences thus appear readily masked by strong task knowledge. We next tested whether sign- and goal-tracking can be assessed online. To this end, we compared five potential measures of sign- and goal-tracking in 125 participants: webcam eye-tracking, mouse-tracking, side-tracking, a dot-probe task, and the Piggy-Game, a gamified task in which participants click on the cue or the reward location. Only for the Piggy-Game, sign-tracking was associated with enhanced PIT, making it the most promising online tool.

T17 Enhanced Pavlovian-to-instrumental transfer in human sign-trackers: bias or slower updating?

Romana Caposova

HMU Health and Medical University, Potsdam, Germany

Reward-predictive cues influence instrumental behaviour more strongly in human sign-trackers than in goal-trackers. This enhanced Pavlovian-to-instrumental transfer (PIT) could reflect stronger cue influence from the outset, slower updating during transfer, or both. Distinguishing these explanations requires examining how the group difference develops across trials. We address this question by reanalysing PIT data from 128 participants reported by Schad et al. (2020). During transfer, participants performed previously learned instrumental responses in the presence of Pavlovian cues predicting monetary outcomes. Outcomes were hidden during PIT, while earnings continued to count towards payment, constituting nominal extinction. We used trial-level generalized linear mixed-effects analyses to separate the initial group difference in the influence of cue value from its subsequent change. Sign-trackers showed a stronger cue-value effect already initially, reflecting enhanced PIT in sign- compared to goal-trackers, but there was no significant change in this group-difference across trials. AIC comparison favoured a model retaining the initial difference without an additional linear change in that difference over trials. These findings support a bias explanation of enhanced PIT in sign-trackers, without establishing differential learning rates. We discuss how these findings may relate to prior evidence for slower updating in sign-trackers, considering the role of feedback availability and task structure.

T18 Sign- and goal-tracking in human alcohol dependence

Nassim Sadedin

HMU Health and Medical University, Potsdam, Germany

Background: Many addictive drugs are thought to control intake via Pavlovian cues. Indeed Pavlovian-instrumental transfer (PIT) is enhanced in dependent patients. PIT is also enhanced in "sign-tracker" individuals, who approach cues during Pavlovian conditioning using model-free learning, whereas "goal-tracker" individuals, who approach the reward location using model-based learning, show reduced PIT. Addiction may relate to model-free sign-tracking, as was found in rodents. However, whether sign- versus goal-tracking underlies phases of human substance dependence is untested.

Methods: We assessed a PIT task, using eye-tracking and fMRI during Pavlovian

conditioning to study human sign- and goal-tracking, and fMRI during the transfer part, in 61 recently detoxified alcohol-dependent patients and 53 matched healthy controls.

Results: The neural and behavioral distinction between sign- and goal-trackers was replicated in the patient group, with sign-trackers relying more on model-free and goal-trackers more on model-based control, visible in behavioral and neural PIT computational modeling of gaze, and in pupillometry. However, against expectations, a) this distinction was not replicated in healthy controls; b) sign-tracking was not increased in patients compared to healthy controls overall, and c) goal-trackers were more likely to relapse. There were multiple age-dependent effects.

Conclusions: Long-term alcohol use does not appear to substantially increase sign-tracking, and sign-tracking itself is not predictive of relapse. Age-dependent changes in sign-/ goal-tracking balance offset by alcohol consumption may explain part of the pattern.

POSTERS

Poster presentations

Poster session

Saturday, 17:15–18:45

P1 Pavlovian bias on decision-making as a key marker distinguishing human sign-trackers from goal-trackers: a comprehensive meta-analysis
Marco Badioli p. 25

P2 Beyond the craved substance? Cue-induced shifts in valuation and pupil dilation across the impulsive-compulsive spectrum
Erik Lukas Bode p. 26

P3 Measures of Human Sign- and Goal-Tracking Differ Across Primary and Secondary Rewards
Mariya V. Cherkasova p. 27

P4 Role of the inter-trial interval in the expression of appetitive conditioned responses
François Cinotti p. 28

P5 Of rats and men: Progress in the study of Pavlovian cue processing & risk for substance use disorder in humans
Lora M. Cope p. 29

P6 Behavioral and computational evidence on how Pavlovian cues shape the adaptation to volatility in sign-trackers
Claudio Danti p. 30

P7 Toward a Sex-Informed Model of Sign- and Goal-Tracking
Luigi Albert Enrico Degni p. 31

P8 Chronic Stress and Pavlovian-to-Instrumental Transfer in Alcohol Use Disorder
Samanda Krasniqi p. 32

P9 Implicit Expectations: Neural Evidence for Reward Pavlovian Conditioning in the Absence of Contingency Awareness
Mateo Leganes-Fonteneau p. 33

P10 Value-modulated attentional capture is sensitive to explicit knowledge
Pablo Martínez López p. 34

P11 Moving the CS and reward location improves gaze measures of sign-tracking: a simulation study
Lorenzo Mattioni p. 34

P12 Effects of reward devaluation on value-modulated attentional capture (VMAC) in the Context of Tobacco Dependence
Lana Mrkonja p. 35

P13 Pavlovian Bias or Explicit Strategy? Awareness and Individual Differences in Human PIT
Adrian Nichici p. 36

ABSTRACTS

Poster abstracts

P1 Pavlovian bias on decision-making as a key marker distinguishing human sign-trackers from goal-trackers: a comprehensive meta-analysis

Marco Badioli

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Although animal research links sign-trackers to maladaptive tendencies, including risk-taking, inflexibility, and impulsivity, in contrast to the greater behavioural flexibility and more adaptive responding of goal-trackers, the extent to which these patterns generalise to humans remains unclear, as a comprehensive quantitative assessment of these associations is lacking. To address this gap, the present meta-analysis, which included 25 studies (130 samples), quantifies differences between human sign-trackers and goal-trackers across several psychological and psychophysiological constructs, with the aim of identifying domains for which evidence of dissociation is robust, as well as those for which findings indicate no reliable differences. Results showed that Pavlovian bias on decision-making construct represents the best marker for distinguishing sign-trackers from goal-trackers, taking into account model effect size, precision estimate, and the prediction interval for future studies. Furthermore, higher self-report impulsivity in sign-trackers emerged as the second construct associated with positive and precise effect size estimates, although its prediction interval included effects in the opposite direction. Notably, higher pupil dilation, self-report substance use, and lower behavioural inhibitory control in sign-trackers emerged as other promising domains, highlighting the need for further research to reduce the substantial uncertainty in the estimated between-group differences. In conclusion, while Pavlovian bias in decision-making may reflect the greater persistence of cue-driven responding in sign-trackers compared to goal-trackers, potentially representing the key maladaptive feature characterising sign-trackers, more adequately powered studies examining other positive but still imprecise constructs are needed. To this end, the present meta-analytic effect size estimates may also provide a useful basis for prospective power analyses in future research.

P2 Beyond the craved substance? Cue-induced shifts in valuation and pupil dilation across the impulsive-compulsive spectrum

Erik Lukas Bode

Charité – Universitätsmedizin Berlin, Germany

Craving, a strong desire for a substance or behavior, is central to disorders on the impulsive-compulsive spectrum such as alcohol use disorder (AUD), gambling disorder (GD) and binge eating disorder (BED). Despite its clinical relevance, the neurocomputational mechanisms underlying craving remain poorly understood. Cue-induced craving can be framed as a Pavlovian process in which reward-predictive cues acquire incentive salience and bias instrumental choice, a process that sign-tracking accounts propose varies between individuals. Pupil dilation, an index of noradrenergic arousal, has been linked to craving and may provide a physiological marker of this cue-driven incentive motivation. Using a computational psychiatry approach, we investigate whether cue-induced craving alters value-based decision-making across diagnostic boundaries. We recruited patients with AUD (n = 31), GD (n = 30) and BED (n = 22), as well as healthy controls (n = 76). Participants completed an established paradigm (Konova et al., 2018) in which exposure to a food cue induces craving, and its effect on economic choice is captured with a computational utility model. Because the food cue is disorder-specific for BED but not for AUD or GD, the design allows us to test whether craving-related shifts in valuation generalize beyond the craved substance. Pupil responses were recorded throughout the task. At the conference, we will present first results from linear mixed-effects models of cue-induced changes in bidding behavior and subjective valence ratings, and their relationship to pupil dilation across groups. This approach aims to clarify shared mechanisms of cue-induced craving across disorders and may help inform targeted interventions.

P3 Measures of Human Sign- and Goal-Tracking Differ Across Primary and Secondary Rewards

Mariya V. Cherkasova

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Background: Sign- and goal-tracking are behavioral phenotypes thought to reflect trait-like differences in the attribution of incentive salience to reward-predictive cues. In rodent Pavlovian conditioning paradigms, sign-trackers approach the reward-predictive cue, whereas goal-trackers approach the site of reward (usually food), typically producing a bimodal distribution of Pavlovian conditioned approach (PCA). Rodent-to-human translations have often operationalized PCA as visual attention to sign versus goal locations. Importantly, in human paradigms, unconditioned stimuli (US) are typically symbolic indicators of secondary (monetary) or primary (e.g. food) rewards that are not immediately available. The translational validity of such tasks is limited by the absence of immediately available primary rewards and resulting PCA distributions that are unimodal and often skewed, challenging the practice of categorizing individuals as sign- versus goal-trackers. We compared PCA using a primary (erotica) versus secondary (monetary) rewards as US in humans.

Methods: Fifty-five non-clinical volunteers completed a Pavlovian conditioning paradigm in which distinct abstract cues served as conditioned stimuli and predicted erotic images, monetary values, or neutral outcomes (scrambled erotic images) presented in the opposite hemifield. PCA was quantified using eye-tracking to measure visual attention to sign versus goal locations during cue presentation, yielding a gaze index from -1 (goal-tracking) to +1 (sign-tracking).

Results: None of the US conditions produced significantly non-unimodal gaze index distributions. However, the erotic stimulus distribution showed a bimodal-resembling shape and was the only distribution that significantly deviated from normality. Erotic US produced significantly greater goal-tracking than monetary and neutral US, which did not differ from each other. The classification of participants as sign- versus goal-trackers varied depending on whether the erotic or monetary US was used, with 25% classified differently using a median split and 20% using a tertile split of the gaze index distribution. Goal-tracking for the erotic US was associated with self-reported sexual desire, supporting a link between goal-tracking and incentive salience of the US. Sign- and goal-tracking in the context of the monetary US was unrelated to self-reported desire for sex or money.

Conclusion: Primary reward US produced a PCA distribution more closely resembling

those observed in rodents. The type of reward used as the US may therefore be an important consideration for the translational validity of human sign- and goal-tracking paradigms.

P4 Role of the inter-trial interval in the expression of appetitive conditioned responses

François Cinotti

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Considerable attention has been devoted to individual differences in the expression of Pavlovian appetitive conditioning in rats. Whereas sign-trackers direct their conditioned responses toward the conditioned stimulus (CS) itself, goal-trackers approach the location of unconditioned stimulus (US) delivery. These behavioral differences have often been associated with stable individual traits or genetic predispositions. However, contextual factors, including the inter-trial interval (ITI), can also influence the expression of sign-tracking and goal-tracking behavior. One possible explanation is that goal-tracking responses undergo extinction between trials. To test this hypothesis, we manipulated contextual cues during the ITI. The introduction of additional cues transiently increased goal-tracking responses during CS presentation but had no effect on sign-tracking responses, consistent with the context-dependent recovery of previously attenuated goal-tracking responses. Moreover, all animals initially exhibited goal-tracking behavior but progressively developed a sign-tracking phenotype over the course of the experiment. These findings highlight the importance of contextual and temporal factors in shaping the expression of Pavlovian conditioned responses and suggest that sign-tracking and goal-tracking phenotypes may evolve with experience.

P5 Of rats and men: Progress in the study of Pavlovian cue processing & risk for substance use disorder in humans

Lora M. Cope

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Background: An exciting question in substance use research is whether ground-breaking preclinical discoveries with respect to sign- and goal-tracking (ST/GT) behaviors and risk for substance misuse replicate in humans. Growing interest in how to reliably measure ST/GT in human participants has spurred the development of novel behavioral paradigms to characterize individual differences in these phenotypes. Here we report on the development of two Pavlovian conditioned approach (PavCA) paradigms: a physical interactive task and a neuroimaging-compatible computerized task.

Methods: Sample 1 (physical interactive task; $N = 54$): Each trial began with extension of a lever for 3 sec, during which participants could press it as much as they wanted. After retraction, a token was immediately delivered into a reward magazine on 1/3 of trials. Metrics of interest are participants' eye-gaze (location, latency, probability) and physical interaction (count, latency, and probability of lever presses and magazine entries) while the lever is extended. Sample 2 (computerized task; $n = 33$, subset of Sample 1): Participants first saw one of two rectangular fractals in the upper left or upper right part of the screen (side counterbalanced across participants). One of the fractal types was rewarded at 67% contingency and the other fractal type was never rewarded. The two fractal types were each presented on 50% of trials. In a rewarded trial, immediately after the fractal disappeared a dollar bill image appeared in the lower center of the screen. Participants pressed a button while the bill was on the screen to "collect" it. Fixation crosses marking the locations of the two fractals and dollar bill were visible on the screen whenever their corresponding stimulus was not, including during the inter-trial interval. Metrics of interest are participants' eye-gaze (location, latency, and probability) while the fractals are on the screen.

Results: In Sample 1, greater ST was associated with cognitive complexity impulsivity ($\tau = .21$, $p = .035$), alcohol craving ($\tau = .38$, $p = .024$), number of black-outs ($\tau = .37$, $p = .050$), and in females only, number of binge-drinking episodes ($\tau = .38$, $p = .021$). In Sample 2, main effects of the reward cue vs. control cue contrast showed engagement of areas associated with cue reactivity, emotional salience, and subjective valuation of potential rewards (precuneus) as well as emotion regulation, reappraisal, and goal-directed behavior (ventrolateral prefrontal cortex). Greater ST was also associated with motor impulsivity ($r = .38$, $p = .007$).

Conclusions: Preliminary findings are promising and support the idea that ST/GT are real constructs that can be reliably measured in humans. Further, results provide evidence that ST/GT are useful for understanding vulnerability to substance use in humans. Our lab has collected over 400 PavCA sessions, and new results—which include ST/GT associations with functional neuroimaging and longitudinal substance use patterns—will be forthcoming.

P6 Behavioral and computational evidence on how Pavlovian cues shape the adaptation to volatility in sign-trackers

Claudio Danti

University of Bologna, Italy

Natural environments are often characterized by inherent uncertainty. While humans learn instrumental contingencies to maximize rewards, they must also be ready for sudden contingency shifts. Volatility refers to how fast the contingencies of an environment change over time. Humans can adapt their learning rate to these changes, but as volatility increases, the precision of goal-directed control is reduced. Recent evidence shows that this makes behavior more open to Pavlovian biases, i.e. learned cue-reward associations that affect action selection independently of current outcomes, with people relying more on Pavlovian cues when the environment is highly volatile. What remains unclear is whether everyone is affected in the same way. Here we ask whether sign-trackers, who attribute strong incentive salience to cues, and goal-trackers, who focus more on the outcome, differ in how flexibly they adjust the use of Pavlovian cues to the volatility of the environment.

We tested this using a Pavlovian-to-Instrumental Transfer paradigm. First, participants learned instrumental action-outcome contingencies in two reversal learning conditions, i.e., under low volatility (one reversal per block) and high volatility (five reversals per block). Then, they acquired Pavlovian cue-outcome associations while their eye movements were recorded, and gaze behavior toward the cues and the outcome location was used to classify them as sign-trackers or goal-trackers. Finally, in the transfer phase, they performed both reversal learning conditions in the presence of the previously learned Pavlovian cues. Preliminary data are consistent with our preregistered hypothesis that the influence of the cues differs between phenotypes as a function of volatility. Sign-trackers followed the cues in both contexts, whereas goal-trackers used them mainly when the environment was volatile.

We propose that, in the transfer phase, the presence of Pavlovian cues makes sign-

trackers lose the adaptation to volatility that is specific to each context, so that they behave in a similar way in stable and volatile environments. Goal-trackers, instead, would keep this adaptation and adjust the influence of the cues to the volatility of the environment. We test this idea with Rescorla-Wagner reinforcement learning models fitted to participants' choices, and preliminary results point in this direction. Overall, these preliminary findings suggest that differences between sign-trackers and goal-trackers may reflect how flexibly they combine Pavlovian and instrumental information when the environment changes.

P7 Toward a Sex-Informed Model of Sign- and Goal-Tracking

Luigi Albert Enrico Degni

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Recent studies in animals and humans have demonstrated sex-associated differences in Pavlovian conditioned approach, with females more likely to exhibit sign-tracking and males more likely to exhibit goal-tracking. Given the theorized link between sign-tracking and maladaptive behaviours, such as impulsivity and addiction, this finding appears difficult to reconcile with evidence indicating higher levels of addiction-related behaviours in males. One possible explanation is that vulnerability to maladaptive behaviour depends on the interaction between sex and Pavlovian conditioning phenotype. Here, I provide preliminary evidence in humans that male sign-trackers exhibit greater suboptimal decision-making and poorer inhibitory control compared to female sign-trackers and both male and female goal-trackers. These findings suggest that the behavioural correlates of sign- and goal-tracking may differ as a function of sex, potentially helping to account for some of the heterogeneity observed in both the sign-/goal-tracking literature and research on sex differences in neuroscience. More broadly, it may represent a first step towards a unified and translational model of sign- and goal-tracking in addiction vulnerability, which should consider sex as a crucial variable for predicting addiction trajectories over time.

P8 Chronic Stress and Pavlovian-to-Instrumental Transfer in Alcohol Use Disorder

Samanda Krasniqi

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Introduction: Alcohol use disorder (AUD) is associated with alterations in motivational learning processes, particularly an enhanced influence of reward-associated cues on behaviour. Pavlovian-to-instrumental transfer (PIT) describes the influence of conditioned stimuli on instrumental choice and is considered a relevant mechanism underlying addiction related behaviour. Chronic stress has further been proposed as a key factor in the development and maintenance of AUD. However, the relationship between chronic stress and PIT remains insufficiently understood. Objectives: This study aimed to investigate whether chronic stress is associated with PIT and whether this relationship differs between individuals with AUD and healthy controls (HC) at the behavioural and neural level. Methods: A total of 307 participants (173 with AUD, 134 HC) were included. Participants in the AUD group met at least two DSM-5 criteria for AUD. Chronic stress was assessed using a multimodal approach comprising hair cortisol, the Trier Inventory for Chronic Stress (TICS) and the Perceived Stress Scale (PSS-10). All participants completed a PIT paradigm during functional magnetic resonance imaging (fMRI), assessing the influence of conditioned cues on instrumental behaviour. Results: Individuals with AUD reported significantly higher perceived stress than HC (PSS-10: $p = .001$) but did not differ in chronic stress as measured by the TICS ($p = .136$) or hair cortisol ($p = .661$). Behavioural analyses revealed a robust PIT effect across both groups ($p < .001$), with no significant difference between AUD and HC. Neither self-reported (PSS 10, TICS) nor biological stress (hair cortisol) moderated the PIT effect or the group difference (all ps > .10). Conclusions: The present behavioural findings do not support a direct association between chronic stress and PIT in AUD. As stress-related processes may be reflected in more subtle neural alterations, differences may emerge at the level of brain activation rather than behaviour. Future analyses of the fMRI data will examine whether chronic stress is associated with the neural correlates of PIT.

P9 Implicit Expectations: Neural Evidence for Reward Pavlovian Conditioning in the Absence of Contingency Awareness

Mateo Leganes-Fonteneau

University of Amsterdam and UCLouvain

Establishing implicit Pavlovian conditioning requires positive evidence both that stimulus–outcome contingencies remained outside awareness and that learning nevertheless occurred. We combined the Bayesian Awareness Categorization Technique with EEG to characterize the neural dynamics of implicit reward conditioning. Participants (n = 66; final analytic sample n = 43) completed a Pavlovian task in which conditioned stimuli predicted high or low probabilities of monetary reward (CS+/CS−) while they performed a distractor task. Sensitive Bayesian evidence was used to classify participants as aware, unaware, or insensitive, and EEG analyses focused on participants classified as unaware. Event-related potentials (P3, CNV, SPN, FRN) and beta-band power were examined across the first and second halves of the task. Cue-evoked P3 amplitudes did not differentiate CS+ from CS− and declined over time. By contrast, SPN amplitudes were greater for CS+ than CS− across both halves, indicating sustained anticipatory differentiation of reward-predictive cues. A complementary anticipatory beta-band effect emerged within the CNV window during the second half, when power was greater for CS+ than CS−. At feedback, FRN amplitudes primarily reflected outcome valence, whereas a frontocentral beta-band cluster during the first half differentiated expected from unexpected outcomes. This temporal dissociation is consistent with a redistribution of expectancy-related processing from feedback to anticipatory stages as learning develops. The findings provide time-resolved neural evidence for reward conditioning in genuinely unaware participants, challenging accounts that regard conscious contingency knowledge as necessary for Pavlovian learning and refining how awareness constrains associative learning.

P10 Value-modulated attentional capture is sensitive to explicit knowledge

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There is plenty of evidence showing that the history of stimulus-reward pairings can cause such stimuli to capture our attention more easily, even when attending to them actually prevents the delivery of a reward. Once established, these Pavlovian attentional biases are believed to operate automatically and persist even when they no longer predict a reward. This reflects an inflexible form of attentional sign-tracking. In this study, we investigated how reversal learning of Pavlovian attentional biases is affected by explicit knowledge. Across three experiments, we demonstrate that verbal instructions alone can modulate previously learned attentional biases (Experiments 1 and 2a) and that, when not informed of the reversal, participants still exhibit strong associations between Pavlovian attentional biases and individual differences in awareness of the new, reversed contingencies (Experiment 2b). Computational modelling of a previously published dataset reveals that instructions may prompt a rapid model-based revaluation by eliciting the automatic updating of the stimuli's value representation. However, such updating can be counterproductive, as attending to the previously high-value distractor reduces the likelihood of receiving a reward when contingencies are re-evaluated. These findings emphasize the importance of explicit knowledge in the updating process.

P11 Moving the CS and reward location improves gaze measures of sign-tracking: a simulation study

Lorenzo Mattioni

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Eye-tracking measures sign- and goal-tracking by comparing looks at the cue (CS) with looks at the reward location. However, in traditional paradigms, both stimuli usually appear at fixed screen locations, which could introduce interference from attentional preferences for certain locations, particularly when stimuli are presented along a vertical axis. Additionally, attempts to account for this by switching locations can generate carry-over effects, increasing noise measurement.

Here we propose a new design in which the CS appears at one of eight positions around a circular stimulus array, locations chosen to prevent carry-over effects across consecutive trials. We simulated 1,000 studies of 60 participants with known sign/goal-

tracking traits, individual spatial biases, and carry-over effects. Three static layouts fixed the CS per participant: up/down, up/down counterbalanced across participants, or left/right. Two dynamic layouts moved the CS either randomly between left and right or among the eight positions. Validity was rank correlation with the true trait.

We observe overall that dynamic layouts improved sign-tracking estimates. Validity was 0.71 for random left/right and 0.73 for eight positions, compared with 0.44–0.55 for traditional static top–bottom layouts. Spatial preferences shifted scores by 0.16–0.25 in these static layouts and by 0.06–0.07 in the dynamic layouts. The two dynamic layouts were almost equal and outperformed fixed layouts in all nine alternative populations.

Subtracting the CS− gaze index from the CS+ index reduced preferences shared by both trial types and raised validity to 0.80–0.85 across layouts. Random left/right was particularly vulnerable to carry-over because the previous reward location could become either the cue or reward location on the next trial. In fixed layouts, carry-over favored the reward location on both CS+ and CS− trials, so subtraction reduced its effect. The eight-position task prevented this mechanism through its rule prohibiting shared positions on consecutive trials. Under strong carry-over, validity fell to 0.71 for random left/right, compared with 0.84–0.86 for the other four layouts.

Eight positions combine dynamic locations with protection against carry-over. Such simulations can guide task design before data collection and help compare results across layouts.

P12 Effects of reward devaluation on value-modulated attentional capture (VMAC) in the Context of Tobacco Dependence

Lana Mrkonja

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In animal models of sign-tracking, cues that predict reward can acquire incentive motivational properties, attracting attention and approach even when they are not themselves rewarding. Value-modulated attentional capture (VMAC) paradigms may assess a related process in humans: stimuli that signal highly valued outcomes come to capture attention automatically, despite being task-irrelevant. However, most VMAC research so far has used monetary rewards as outcomes. The present study examines whether cues predicting substance-related rewards similarly acquire attentional priority, and whether such attentional capture remains sensitive to changes in the current motivational value of the predicted outcome.

Light and heavy smokers complete a VMAC task in which task-irrelevant distractor colors predict rewards in either a tobacco-related or monetary reward condition. In tobacco-related trials, distractor color signals the availability of either a high-value tobacco reward or a low-value water reward. In monetary trials, distractor color signals either a high (10 cents) or low (1 cent) monetary reward. Following initial assessment of attentional capture, participants undergo a tobacco-reward devaluation manipulation before completing the VMAC task again. This provides a test of whether attention to tobacco-predictive cues tracks the current motivational value of the tobacco outcome or persists despite devaluation.

Preliminary findings indicate greater attentional capture by cues signaling high- compared with low-value outcomes in both the tobacco-related and monetary reward conditions. Importantly, attentional capture by tobacco-predictive cues persisted following tobacco-reward devaluation. Thus, once acquired, attentional priority for substance-reward cues appeared relatively resistant to changes in the current motivational value of the predicted reward, consistent with relatively automatic or "habit-like" behavior. This provides a potential human attentional analogue of the persistent sign-tracking behavior observed in animals. However, the current sample consists primarily of light or social smokers, and the relatively small number of more dependent smokers limits conclusions regarding individual differences in addictive behaviour and nicotine dependence.

P13 Pavlovian Bias or Explicit Strategy? Awareness and Individual Differences in Human PIT

Adrian Nichici

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Previous studies have suggested that Pavlovian-to-instrumental transfer (PIT) requires participants' awareness of Pavlovian contingencies. However, these studies have often used relatively simple Pavlovian learning procedures, resulting in high rates of explicit contingency knowledge. Further, unawareness has usually been inferred from non-significant contingency awareness tests, although a null effect in such tests does not provide evidence for the absence of awareness.

In this study, we addressed these limitations by using an adapted PIT paradigm together with a Bayesian awareness classification method that allowed us to distinguish between evidence for awareness, evidence for unawareness, and inconclusive evidence. We recruited 80 participants, of whom 15 were classified as aware, 48 as unaware, and 17 as

inconclusive.

We observed interference PIT in participants classified as aware, but not in participants classified as unaware. Critically, the effect in the aware group was driven by a small number of participants who responded almost deterministically according to the Pavlovian contingencies. In these participants, the Pavlovian cues did not appear to merely bias responding but acted in manner consistent with an explicit rule-like strategy. Qualitatively similar patterns of strong between-subject PIT variability, including bimodal PIT distributions, have been reported previously, suggesting that group-level PIT effects may sometimes be disproportionately influenced by participants responding to Pavlovian cues in a rule-like fashion.

These findings provide no evidence for PIT in the absence of contingency awareness and raise the possibility that human PIT effects may at least partly be driven by the explicit use of Pavlovian contingency knowledge rather than a purely implicit biasing influence of Pavlovian cues. Distinguishing these mechanisms may be crucial for interpreting individual differences in PIT and their associations with psychopathology.

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See you in Potsdam!

Thank you for being part of this workshop. We hope for lively discussions, new ideas and good conversations over coffee.

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